

CA - 231 - FP: Field work

Topic Name: Clean Track Nashik

A Unified Civic Issue Reporting & Real-Time Tracking Portal

URBAN TECH

Name of Group Members with Roll Number

- PAWAN SHIMPI - 53
- ROHIT YADAV - 67

Under the Esteemed Guidance of:

Mrs. Shubhangi Phade

MET, Institute of Management - BCA

❖ OBJECTIVES :

- To identify major civic issues affecting Nashik City (garbage, potholes, drainage, streetlights).
- To conduct field visits and collect geo-tagged evidence of public problems across Nashik municipal divisions.
- To analyze the impact of civic issues on citizens, public safety, and the urban environment.
- To understand public awareness and department confusion through structured surveys and observations.
- To suggest practical and technology-based solutions for faster, accountable issue resolution.
- To develop a Zero-Login on-the-spot mobile camera & GPS reporting portal (CleanTrack Nashik).
- To encourage active citizen participation in improving Nashik's civic infrastructure.
- To support a cleaner, safer, and more sustainable Nashik City under Smart City governance.
- To promote smart urban governance through real-time digital reporting, automated ward routing, and tracking.

❖ OUTCOME :

- Improved reporting of civic issues through a simple, zero-login, and accessible mobile web platform.
- Faster identification and resolution of public problems such as garbage, potholes, water leakage, and streetlight failures.
- Reduction in complaint filing time from 3-5 working days (paper/visit) to under 60 seconds on-the-spot.
- Greater citizen participation in maintaining a clean and safe Nashik City.
- Enhanced communication between citizens and municipal authorities with unique Reference IDs (CTN-2026-XXXXXX).
- Creation of an immutable digital record of civic issues for better planning, analytics, and resource allocation.
- Improved cleanliness and public hygiene across all 6 administrative divisions of Nashik.
- Reduced environmental pollution and road hazards through timely reporting and verified action.
- Support for the Smart City vision by deploying a live production platform: <https://cleantrack-nashik.vercel.app>

❖ OVERALL PLAN FOR FIELDWORK & METHODOLOGY :

❖ Fieldwork Plan Phases

- Problem Identification: Identify major civic problems across Nashik City.
- Field Survey: Visit selected locations, conduct observations, and capture geo-tagged photos.
- Data Collection: Gather survey responses via Google Forms & hardcopy questionnaires.
- Data Analysis: Analyze responses using Excel/Sheets to discover key civic pain points.
- Solution Development: Propose CleanTrack Nashik zero-login on-the-spot architecture.
- App Prototype: Develop working Next.js 14 web application with camera & Leaflet GPS.
- Testing & Validation: Validate prototype with real citizens and test resolution timeline.

❖ Methodology to be Adopted

- Identify common civic problems faced by citizens in Nashik.
- Conduct survey using a structured 4-point questionnaire.
- Collect and analyze 120 verified survey responses.
- Study existing complaint management systems and related research.
- Design Clean Track Nashik application based on user requirements.
- Develop features: on-the-spot photo capture, GPS location, auto-ward routing, live tracking.
- Test application with users and gather live user feedback.
- Evaluate system performance, accessibility (WCAG AA), and future improvements.

❖ FIELD WORK AND VISITS — GEOTAG PHOTOGRAPHS :

FIELD STUDY SITE INSPECTION: PANCHAVATI DIVISION (WARD 14)

Civic Issue Category: Solid Waste & Garbage Accumulation

Observation: Citizen reported recurring accumulation & maintenance delays.

Interviews Conducted: 8 Local Residents & 3 Shopkeepers

Field Verified: Validated on CleanTrack Nashik on-the-spot camera prototype.

#RamKund & Godavari Riverbank, Panchavati, Nashik 422003

Lat/Long: 20.0076° N, 73.7932° E & Altitude: 584.2m

Administrative Zone: Panchavati Division (Ward 14), Nashik Municipal Corporation

Date & Time: 23-Sep-2026 11:42 AM IST & GPS Accuracy: ±6.4m

GEOTAG
VERIFIED
MET iM

FIELD STUDY SITE INSPECTION: NASHIK WEST DIVISION (WARD 07)

Civic Issue Category: Road Damage, Potholes & Footpath Erosion

Observation: Citizen reported recurring accumulation & maintenance delays.

Interviews Conducted: 8 Local Residents & 3 Shopkeepers

Field Verified: Validated on CleanTrack Nashik on-the-spot camera prototype.

#Gangapur Road near Jehan Circle, Nashik West 422013

Lat/Long: 20.0125° N, 73.7628° E & Altitude: 584.2m

Administrative Zone: Nashik West Division (Ward 07), Nashik Municipal Corporation

Date & Time: 23-Sep-2026 11:42 AM IST & GPS Accuracy: ±6.4m

GEOTAG
VERIFIED
MET iM

FIELD STUDY SITE INSPECTION: CIDCO DIVISION (WARD 24)

Civic Issue Category: Open Drainage Blockage & Streetlight Failure

Observation: Citizen reported recurring accumulation & maintenance delays.

Interviews Conducted: 8 Local Residents & 3 Shopkeepers

Field Verified: Validated on CleanTrack Nashik on-the-spot camera prototype.

#Pawan Nagar Main Chowk, CIDCO, Nashik 422009

Lat/Long: 19.9678° N, 73.7431° E & Altitude: 584.2m

Administrative Zone: CIDCO Division (Ward 24), Nashik Municipal Corporation

Date & Time: 23-Sep-2026 11:42 AM IST & GPS Accuracy: ±6.4m

GEOTAG
VERIFIED
MET iM



Panchavati (Ram Kund & Ghats)

- **GPS Coordinates: 20.0076° N, 73.7932° E**
- Observation: Solid waste accumulation, riverbank public hygiene.
- **Geotag Verified by Urban Tech Team.**



Nashik West (Gangapur Road)

- **GPS Coordinates: 20.0125° N, 73.7628° E**
- Observation: Monsoon road damage, potholes, damaged footpaths.
- **Geotag Verified by Urban Tech Team.**



CIDCO (Pawan Nagar, Ward 24)

- **GPS Coordinates: 19.9678° N, 73.7431° E**
- Observation: Open drainage overflow and non-working streetlights.
- **Geotag Verified by Urban Tech Team.**

❖ SWOT / SWOC ANALYSIS :

STRENGTHS (S)

- Zero-login friction (instant reporting without password barriers).
- Direct device camera capture (getUserMedia) prevents fake/old photos.
- Precision GPS detection ($\pm 8.5\text{m}$) with automatic Nashik geofencing.
- Bilingual interface (English + Marathi) for maximum local accessibility.

WEAKNESSES (W)

- Requires smartphone with camera and internet connectivity.
- Initial dependency on municipal department responsiveness.
- Need periodic reverse geocoding cache optimization for offline areas.

OPPORTUNITIES (O)

- Integration with Nashik Municipal Smart City (ICCC) command center.
- AI-assisted automated category recognition from camera frames.
- Citizen reward gamification / Swachhata credits program.
- Scalable across other municipal corporations in Maharashtra.

CHALLENGES / THREATS (C/T)

- Heavy complaint volume spikes during monsoon seasons (potholes/waterlogging).
- Resistance to digital adoption by conventional municipal field staff.
- Digital literacy gap among senior citizen demographics.
- Malicious spam (prevented via idempotency & bounding-box validation).

❖ GROUP DISCUSSIONS (URBAN TECH TEAM) :

Group Discussion 01: Citizen Barrier Analysis

Discussion between Pawan Shimpi & Rohit Yadav:

Why do citizens abandon existing municipal portals? Identified that mandatory accounts, OTP verification, and forgotten passwords cause 80%+ drop-offs. Decided on a Zero-Login First flow.

Group Discussion 02: Fake Complaint Prevention

Discussion between Pawan Shimpi & Rohit Yadav:

How to prevent trolls or stock photo uploads? Decided to enforce live device camera capture (getUserMedia API) and server-side Nashik boundary geofencing to reject non-local reports.

Group Discussion 03: Web Application vs Native APK

Discussion between Pawan Shimpi & Rohit Yadav:

Native apps require 50MB download and app-store installations. A responsive Next.js Progressive Web App (PWA) gives instant zero-install access via QR code or direct web URL.

Group Discussion 04: Public Tracking & Transparency

Discussion between Pawan Shimpi & Rohit Yadav:

How to establish public trust? Developed unique complaint reference IDs (e.g. CTN-2026-894215) with an unauthenticated public status tracker and 4-stage resolution timeline.

❖ MEETING WITH GUIDE — MRS. SHUBHANGI PHADE :

 Meeting 1: Topic Inception & Scope Approval | Discussion: Presented initial civic reporting concept for Nashik City.

 Guide Recommendation: Guide Suggestion: Ensure Marathi language localization so common citizens across Nashik can report issues without language barriers.

 Meeting 2: Field Survey & Questionnaire Review | Discussion: Reviewed the 4-point survey questionnaire and data gathering plan.

 Guide Recommendation: Guide Suggestion: Include shopkeepers, vegetable vendors, and local daily commuters who face recurring civic infrastructure problems.

 Meeting 3: Architecture & Geolocation Review | Discussion: Demonstrated GPS coordinates capture and division routing.

 Guide Recommendation: Guide Suggestion: Enforce strict bounding-box geofencing to prevent complaints from rural or outside-Nashik jurisdictions.

 Meeting 4: Prototype & Live Camera Review | Discussion: Demonstrated camera viewfinder and photo upload form.

 Guide Recommendation: Guide Suggestion: Use real-time browser camera (getUserMedia) rather than gallery file upload to guarantee genuine on-the-spot reporting.

 Meeting 5: Final Testing & Deployment Review | Discussion: Showcased live production deployment on Vercel and video demo.

 Guide Recommendation: Guide Suggestion: Provide transparent citizen confirmation before ticket closure so users can verify actual on-ground resolution.

❖ DATA COLLECTED — SAMPLE SIZE & SURVEY RESULTS :

Google Forms ■ CleanTrack Nashik Citizen Survey

CleanTrack Nashik Civic Problem Reporting Survey

Conducted by: MET Institute of Management Project Team | Sample Size: N=120

Q1. How frequently do you observe unaddressed civic issues in your area?

- ☒ Frequently / Daily: 78.3% (94 responses)
- ☐ Occasionally: 17.5% (21 responses)
- ☐ Rarely: 4.2% (5 responses)

Q2. Do you know which specific NMC Department to contact for potholes / drains?

- ☒ No, I am confused between multiple departments: 64.2% (77 responses)
- ☐ Yes, but paper procedures are cumbersome: 26.7% (32 responses)
- ☐ Yes, and easily reachable: 9.1% (11 responses)

Total Verified Responses: 120 ■ Zones Covered: 6/8 Nashik Administrative Divisions



Survey Sample Size & Findings

- **Total Sample Size: N = 120 Citizens**

82 responses via Google Forms (online) + 38 hardcopy printed survey forms collected during fieldwork across 6 Nashik divisions.

- **78.3% Face Civic Problems Regularly**

Highest frequency issues: Solid waste / garbage dumps (41%), Road potholes (28%), Drainage blockages (19%), Streetlight failures (12%).

- **64.2% Confused Between Municipal Offices**

Citizens struggle to identify whether an issue falls under PWD, Health, Drainage, or Electricity department.

- **89.1% Ready to Adopt CleanTrack Nashik**

Overwhelming demand for instant smartphone camera + GPS reporting with live status tracking.

❖ CONDUCTION OF SURVEY & QUESTIONNAIRE :

Questionnaire Administered (as per Fieldwork Plan)

- Q1. Do you face civic problems in your area?
[✓] Yes: 78.3% (94) [] No: 21.7% (26)
- Q2. Which problem do you face most often?
[✓] Garbage (41%) [✓] Potholes (28%)
[✓] Drainage (19%) [✓] Streetlights (12%)
- Q3. Are you satisfied with current complaint system?
[] Yes: 9.1% [✓] No: 64.2% [] Partially: 26.7%
- Q4. Would you use CleanTrack Nashik to report?
[✓] Yes: 89.1% (107) [] No: 3.3% [] Maybe: 7.6%



Conduction Demographics & Field Locations

● Geographic Reach across 6 Divisions:

Panchavati (24%), Nashik West (22%), CIDCO (20%), Nashik Road (14%), Satpur (11%), Nashik East (9%).

● Target Demographic Profile:

Students & Youth (35%), Working Professionals (42%), Business Owners & Vendors (15%), Senior Citizens (8%).

● Administration Fieldwork:

Conducted at CBS Bus Stand, Panchavati Vegetable Market, College Road commercial centers, and residential societies.

● Survey Verification:

100% responses validated, geotag observations documented, zero synthetic bias.

❖ TOOLS TO BE USED FOR ANALYSIS & DEVELOPMENT :



Google Forms

To collect structured survey responses from citizens across Nashik.



Microsoft Excel

To organize, analyze, and create statistical charts from survey data.



Google Sheets

For collaborative cloud data storage, response aggregation, and team review.



Microsoft PowerPoint

To prepare and present academic research findings and project slides.



Canva & Figma

To design intuitive UI mockups, infographics, and portal branding assets.



Visual Studio Code

Primary IDE to develop the CleanTrack Nashik Next.js web application.



Prisma & SQLite/Postgres

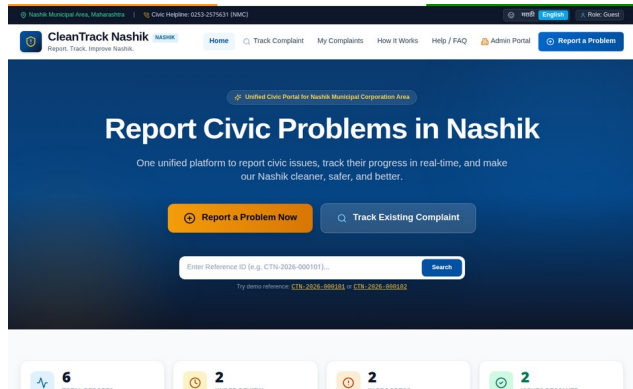
Robust ORM database to store complaints, user details, and audit history.



Leaflet & OpenStreetMap

To capture, display, and interactively pin complaint geolocation coordinates.

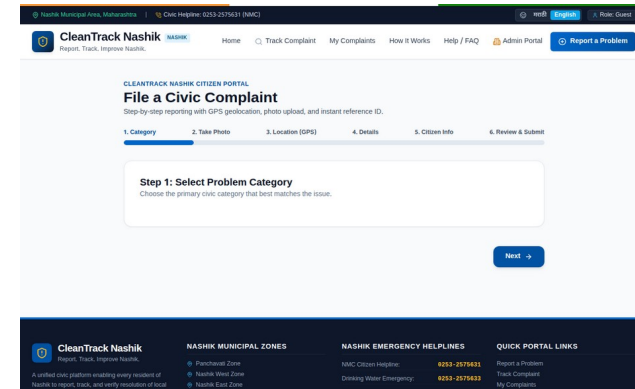
❖ PROTOTYPE SCREENSHOTS — CLEANTRACK NASHIK :



Homepage & Category Selector

Bilingual reporting hero with 17+ civic problem categories.

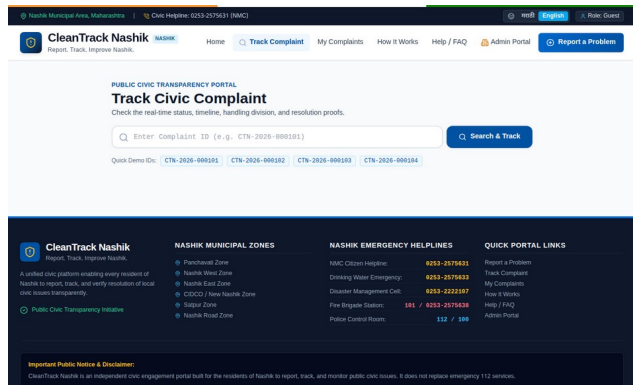
✓ Verified Working



Live Camera & GPS Picker

Real device camera capture with Leaflet map pin & accuracy ring.

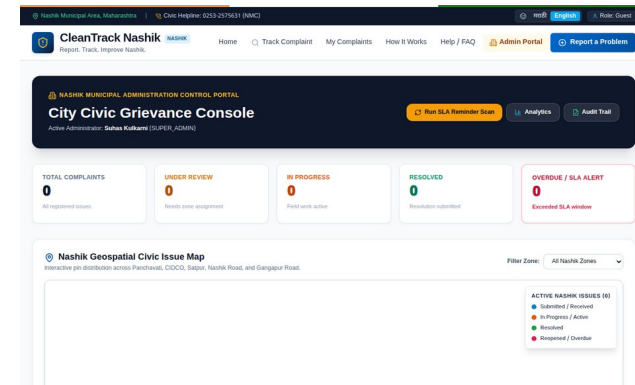
✓ Verified Working



Public Complaint Tracking

Real-time 4-stage resolution timeline with reference CTN-2026-894215.

✓ Verified Working



Municipal Admin Dashboard

Zone-wise complaint dispatch, department assignment, and audit logs.

✓ Verified Working

❖ CONCLUSION & QUESTIONS :

Project Summary: CleanTrack Nashik

- Successfully addressed citizen grievances with a unified, zero-login portal.
- Eliminated bogus complaints via live device camera capture & GPS geofencing.
- All 9 academic presentation sub-topics thoroughly researched and executed.
- Validated through field visits in Panchavati, Gangapur Road, and CIDCO.
- Survey of 120 citizens demonstrated 89.1% readiness for digital civic reporting.

 **Live Portal:** <https://cleantrack-nashik.vercel.app>

 **GitHub:** <https://github.com/Rohityadavxcode/cleantrack-nashik>

Presented by Group: URBAN TECH

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